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A	 PT. ABB Sakti Industri Jalan Gajah Tunggal km 1.0 Tangerang 15135, Indonesia								A																																												
B									B																																												
C	Project Name : MV Switchgear for MNA Serang Plant Switchgear Name : MVD-31 20kV Texturing Plant Panel Type : UNISEC Customer Name : PT Multimas Nabati Asahan Contract No. : BE318042 TITLE : SLD & GA Drawing								C																																												
D									D																																												
E									E																																												
F	<table><tr><td></td><td></td><td></td><td></td><td>Date Prep.</td><td>15.11.2018</td><td>PROJECT TITLE</td><td rowspan="3"> ABB EPMV PT. ABB Sakti Industri </td><td rowspan="3">COVER SHEET SLD & GA Drawing MVD-31 20kV Texturing Plant</td><td>Scale</td><td></td><td>= H00</td></tr><tr><td>1</td><td>Issued for Approval</td><td>18.12.2018</td><td>YR</td><td>Prepared</td><td>YR</td><td>MV Switchgear for MNA Serang Plant</td><td>Customer</td><td>PT Multimas Nabati Asahan</td><td>& GAD</td></tr><tr><td>0</td><td>First Issued</td><td>19.11.2018</td><td>YR</td><td>Checked</td><td>AW</td><td></td><td>Drawing Nbr.</td><td></td><td>Sh. 1</td></tr><tr><td>Ind.</td><td>Revision</td><td>Date</td><td>Name</td><td>Approved</td><td>AI</td><td>Based on:</td><td>BE318042</td><td></td><td>Customer Doc. Nbr.</td><td></td><td>Cont. 17</td></tr></table>												Date Prep.	15.11.2018	PROJECT TITLE	ABB EPMV PT. ABB Sakti Industri	COVER SHEET SLD & GA Drawing MVD-31 20kV Texturing Plant	Scale		= H00	1	Issued for Approval	18.12.2018	YR	Prepared	YR	MV Switchgear for MNA Serang Plant	Customer	PT Multimas Nabati Asahan	& GAD	0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.		Sh. 1	Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042		Customer Doc. Nbr.		Cont. 17	F
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A	SHEET INDEX								A				
B	PAGE		PAGE NAME		REVISION								
	==MVD-31=H00&GAD/1		COVER SHEET		0	1							
	==MVD-31=H00&GAD/2		INDEX OF SHEETS		0	1							
	==MVD-31=H00&GAD/3		GENERAL CHARACTERISTICS		0	1							
	==MVD-31=H00&GAD/4		REFERENCE DESIGNATIONS		0	1							
	==MVD-31=H00&GAD/5		REFERENCE DESIGNATIONS		0	1							
	==MVD-31=H00&GAD/6		REFERENCE DESIGNATIONS		0	1							
	==MVD-31=H00&GAD/7		REFERENCE DESIGNATIONS		0	1							
	==MVD-31=H00&GAD/8		SINGLE LINE DIAGRAM		0	1							
	==MVD-31=H00&GAD/9		SINGLE LINE DIAGRAM		0	1							
	==MVD-31=H00&GAD/10		INSTALLATION RULES		0	1							
	==MVD-31=H00&GAD/11		FRONT VIEW & GA		0	1							
	==MVD-31=H00&GAD/12		FOUNDATION FRAMES		0	1							
	==MVD-31=H00&GAD/13		FOUNDATION FRAMES DETAILS		0	1							
	==MVD-31=H00&GAD/14		FOUNDATION FRAMES DETAILS		0	1							
	C	==MVD-31=H00&GAD/15		SECTION VIEW		0	1						
==MVD-31=H00&GAD/16		SECTION VIEW		0	1								
==MVD-31=H00&GAD/17		SECTION VIEW		0	1								
D									D				
E									E				
F					Date Prep.	15.11.2018	PROJECT TITLE						
	1	Issued for Approval	18.12.2018	YR	Prepared	YR	MV Switchgear for MNA Serang Plant		ABB EPMV	INDEX OF SHEETS SLD & GA Drawing MVD-31 20kV Texturing Plant	Scale		= H00
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A	CHARACTERISTICS				NOTES				A																																																																																
	(IN COMPLIANCE WITH STANDARD IEC 62271-200)																																																																																								
B	SWITCHGEAR VERSION				= UNISEC				B																																																																																
	RATED VOLTAGE (Un)				= 24 kV																																																																																				
C	OPERATING VOLTAGE				= 20 kV				C																																																																																
	RATED FREQUENCY (fr)				= 50 Hz																																																																																				
D	RATED LIGHTNING IMPULSE WITHSTAND VOLTAGE (Up)				= 125 kV				D																																																																																
	RATED POWER FREQUENCY WITHSTAND VOLTAGE (Ud)				= 50 kV																																																																																				
E	RATED CURRENT OF MAIN BUSBARS (Ir)				= 630 A				E																																																																																
	RATED SHORT-TIME WITHSTAND CURRENT (Ik)				= 20 kA																																																																																				
F	RATED PEAK WITHSTAND CURRENT (Ip)				= 50 kA				F																																																																																
	RATED DURATION OF SHORT CIRCUIT (tk)				= 1 s																																																																																				
G	INTERNAL ARC CLASSIFICATION (IAC)				= AFLR / 16kAx1s				G																																																																																
	AMBIENT CONDITION				= NORMAL																																																																																				
H	AMBIENT AIR TEMPERATURE				= -5°C...+40°C				H																																																																																
	DEGREE OF PROTECTION (OPERATION SEATS EXCLUDED)				= IP3X																																																																																				
I	DEGREE OF PROTECTION WITH OPEN DOORS				= IP2X				I																																																																																
	RATED SUPPLY VOLTAGE OF CONTROL AND SIGNALLING CIRCUITS (Ua)				= 230VAC																																																																																				
J	RATED SUPPLY VOLTAGE OF SPRING CHARGING MOTOR (Ua)				= 230VAC				J																																																																																
	RATED SUPPLY VOLTAGE OF LIGHTING AND HEATING CIRCUITS (Ua)				= 230VAC																																																																																				
K	TYPE OF CABLE FOR LOW VOLTAGE CONDUCTORS				= 450/750V				K																																																																																
	TYPE OF CABLE FOR LOW VOLTAGE CONDUCTORS				= PVC																																																																																				
L	CROSS-SECTION OF CONDUCTORS FOR CURRENT CIRCUITS				= 2.5 mm²/BLACK				L																																																																																
	CROSS-SECTION OF CONDUCTORS FOR VOLTAGE CIRCUITS				= 1.5 mm²/BLACK																																																																																				
M	CROSS-SECTION OF CONDUCTORS FOR MOTORS				= 2.5 mm²/BLACK				M																																																																																
	CROSS-SECTION OF OTHER CONDUCTORS (CONTROL AND SIGNALLING CIRCUITS)				= 1.5 mm²/BLACK																																																																																				
N	CROSS-SECTION OF CONDUCTORS FOR INTERCONNECTIONS				= 1.5 mm²/BLACK				N																																																																																
	CROSS-SECTION OF CONDUCTORS FOR INTERCONNECTIONS OF SUPPLY VOLTAGE				= 2.5 mm²/BLACK																																																																																				
O	COMMUNICATION CABLE				= N/A				O																																																																																
FOR IDENTIFICATION OF CONDUCTORS THE "LOCAL END CONNECTION LABELLING" SYSTEM IS USED, IN COMPLIANCE WITH STANDARD IEC 62491 PARAGRAPH 6.2. THIS SYSTEM FORESEES THAT THE MARKING OF A CONDUCTOR END SHOWS THE DESIGNATION OF THE TERMINAL TO WHICH IT IS CONNECTED. IN COMPLIANCE WITH STANDARD IEC 61666 EACH CONDUCTOR IS IDENTIFIED WITH THE REFERENCE DESIGNATION OF THE CONNECTED ELECTRICAL COMPONENT FOLLOWED BY SIGN ":" (COLON), FOLLOWED BY THE TERMINAL DESIGNATION. FOR EXAMPLE THE CONDUCTOR CONNECTED TO TERMINAL "10" OF THE COMPONENT WITH DESIGNATION "-XA" HAS THE MARKING "-XA:10". THE CONDUCTOR MARKING IS PLACED IN SUCH A WAY TO READ IT ALWAYS FROM THE TERMINAL TOWARDS THE CONDUCTOR, AS SHOWN BELOW																																																																																									
01:VQX- 9 10 11 -XDA -XDA:10																																																																																									
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A	REFERENCE DESIGNATION OF OBJECTS IN ELECTRICAL DOCUMENTS														A
	(IN COMPLIANCE WITH STANDARD IEC 81346-2 AND ABB TECHNICAL STANDARD 2NBA000001)														
B	DESIGNATION	DESCRIPTION													B
	-AA	MULTIFUNCTION UNIT (CENTRAL UNIT)													
	-BAD1,-BAD4	CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L1													
	-BAD2,-BAD5	CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L2													
	-BAD3,-BAD6	CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L3													
	-BAR	VOLTAGE PROTECTION RELAY													
	-BAS1	VOLTAGE SENSOR LOCATED ON PHASE L1													
	-BAS2	VOLTAGE SENSOR LOCATED ON PHASE L2													
	-BAS3	VOLTAGE SENSOR LOCATED ON PHASE L3													
	-BAT1	VOLTAGE TRANSFORMER LOCATED ON PHASE L1													
C	-BAT2	VOLTAGE TRANSFORMER LOCATED ON PHASE L2													C
	-BAT3	VOLTAGE TRANSFORMER LOCATED ON PHASE L3													
	-BAT4	VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L1													
	-BAT5	VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L2													
	-BAT6	VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L3													
	-BCD	DIFFERENTIAL PROTECTIVE RELAY													
	-BCF	FEEDER PROTECTION RELAY													
	-BCG	GENERATOR PROTECTION RELAY													
	-BCM	MOTOR PROTECTION RELAY													
	-BCN	NEUTRAL (RESIDUAL) CURRENT TRANSFORMER													
D	-BCP	TRANSFORMER PROTECTION RELAY													D
	-BCR	CURRENT PROTECTION RELAY													
	-BCS1	CURRENT SENSOR LOCATED ON PHASE L1													
	-BCS2	CURRENT SENSOR LOCATED ON PHASE L2													
	-BCS3	CURRENT SENSOR LOCATED ON PHASE L3													
	-BCT1	CURRENT TRANSFORMER LOCATED ON PHASE L1													
	-BCT2	CURRENT TRANSFORMER LOCATED ON PHASE L2													
	-BCT3	CURRENT TRANSFORMER LOCATED ON PHASE L3													
	-BCT4,-BCT7	CURRENT TRANSFORMER LOCATED ON PHASE L1													
	-BCT5,-BCT8	CURRENT TRANSFORMER LOCATED ON PHASE L2													
E	-BCT6,-BCT9	CURRENT TRANSFORMER LOCATED ON PHASE L3													E
	-BCZ	DISTANCE PROTECTION RELAY													
	-BEF	FREQUENCY PROTECTION RELAY													
	-BER	SUPERVISION RELAYS													
	-BES	SYNCHRONIZING RELAY													
	-BET	THERMAL PROTECTION RELAY													
	-BGA1	POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN CIRCUIT-BREAKER COMPARTMENT													
	-BGA2	POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN BUSBAR COMPARTMENT (SINGLE BUSBAR SYSTEM) OR IN FRONT BUSBAR COMPARTMENT (DOUBLE BUSBAR SYSTEM)													
	-BGA3	POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN CABLE COMPARTMENT													
	-BGA4	POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN REAR BUSBAR COMPARTMENT (DOUBLE BUSBAR SYSTEM)													
F	-BGB1...-BGB3	POSITION SWITCHES OF CIRCUIT-BREAKER													F
	-BGB4	POSITION SWITCH OF CIRCUIT-BREAKER: PASSING CONTACT CLOSING MOMENTARILY WHEN THE BREAKER OPENS													

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A	-BGI31	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION						-CA	CAPACITORS							A	
	-BGI32	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN CLOSED POSITION						-CA1	CAPACITOR FOR FRONT VENTILATOR								
	-BGI33	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)						-CA2	CAPACITOR FOR REAR VENTILATOR								
	-BGI41...	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN FEEDER POSITION						-EA1	LIGHTING LAMP LOCATED IN L.V. INSTRUMENT COMPARTMENT.								
	-BGI42	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS IN OPEN POSITION						-EA2	LIGHTING LAMP LOCATED IN CABLE COMPARTMENT								
B	-BGI43...	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN EARTH POSITION						-EA3	LIGHTING LAMP LOCATED IN OPERATING MECHANISM ENCLOSURE							B	
	-BGI51	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN FEEDER POSITION						-EA4	LIGHTING LAMP LOCATED IN CIRCUIT BREAKER COMPARTIMENT								
	-BGI52	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 IN OPEN POSITION						-EB1	HEATER LOCATED IN CABLE COMPARTMENT								
	-BGI53	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN EARTH POSITION						-EB2	HEATER LOCATED IN CIRCUIT-BREAKER COMPARTMENT								
	-BGK	POSITION SWITCH OF KEY LOCK						-EB3	HEATER LOCATED IN MOTOR CABINET								
C	-BGL1	POSITION SWITCH OF ELECTROMECHANICAL LOCK -RLE1						-EB4	HEATER FOR VENTILATION TEST							C	
	-BGS1....-BGS4	POSITION SWITCHES OF CIRCUIT-BREAKER SPRINGS						-EB5	HEATER LOCATED IN L.V. INSTRUMET COMPARTMENT.								
	-BGS5	PROXIMITY SWITCH SIGNALLING SPRINGS IN CHARGED POSITION						-EB7	HEATERS LOCATED IN LEFT SIDE OF THE OPERATING MECHANISM ENCLOSURE								
	-BGT1	POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN SERVICE POSITION						-EB8	HEATERS LOCATED IN RIGHT SIDE OF THE OPERATING MECHANISM ENCLOSURE								
	-BGT2	POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN TEST POSITION						-FA1	SURGE ARRESTER LOCATED ON PHASE L1								
D	-BGT3	POSITION SWITCH ON TRUCK SIGNALLING TRUCK NOT IN ISOLATING TRAVEL POSITION						-FA2	SURGE ARRESTER LOCATED ON PHASE L2							D	
	-BGT4	POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN SERVICE POSITION						-FA3	SURGE ARRESTER LOCATED ON PHASE L3								
	-BGT5	POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN TEST POSITION						-FCD	FUSE-DISCONNECTORS FOR PROTECTION OF AUXILIARY CIRCUITS								
	-BGT6	POSITION SWITCHES SIGNALLING V.T. TRUCK IN SERVICE POSITION						-FCF1	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L1								
	-BGT7	POSITION SWITCHES SIGNALLING V.T. TRUCK IN TEST POSITION						-FCF2	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L2								
E	-BGT8	POSITION SWITCHES SIGNALLING EARTHING TRUCK IN SERVICE POSITION						-FCF3	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L3							E	
	-BGT11	PROXIMITY SWITCH SIGNALLING TRUCK IN SERVICE POSITION						-FCM1	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF SPRINGS CHARGING MOTOR ON MAIN CIRCUIT-BREAKER								
	-BGT12	PROXIMITY SWITCH SIGNALLING TRUCK IN TEST POSITION						-FCM2...-FCM10	MINIATURE CIRCUIT-BREAKERS FOR PROTECTION OF AUXILIARY CIRCUITS								
	-BGV1	POSITION SWITCH OF FRONT VENTILATOR						-FCM11	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV1								
	-BGV2	POSITION SWITCH OF REAR VENTILATOR						-FCM12	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV2								
F	-BM	HYGROSTAT						-FCM13	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV3							F	
	-BPA4	PRESSURE SWITCH LOCATED IN CABLE COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-FCT	THERMAL OVERLOAD RELEASES								
	-BPA5	PRESSURE SWITCH LOCATED IN CIRCUIT-BREAKER OR IN UPPER VT COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GA	GENERATORS								
	-BPA6	PRESSURE SWITCH LOCATED IN BUSBAR OR IN BUSBAR AND CIRCUIT-BREAKER COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ1	FRONT VENTILATOR								
	-BPA7	PRESSURE SWITCH LOCATED IN VOLTAGE TRANSFORMER ISOLATING COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ2	REAR VENTILATOR								
								-GQ3	VENTILATOR LOCATED ON CIRCUIT-BREAKER TRUCK								
								-KFA1	AUXILIARY RELAY SIGNALLING LOW GAS PRESSURE								
								-KFA2	AUXILIARY RELAY SIGNALLING INSUFFICIENT GAS PRESSURE								
								-KFA3....-KFA9	AUXILIARY RELAYS OR CONTACTORS								
								-KFC	CLOSING RELAYS OR CONTACTORS								
								-KFI	INTEGRATED CIRCUITS								
								-KFL	LOCKOUT RELAY								
								-KFM	MICROPROCESSORS								
								-KFN	ANTIPUMPING RELAYS								
								-KFO	OPENING RELAYS OR CONTACTORS								
								-KFP	PROGRAMMABLE LOGIC CONTROLLERS (PLC)								
								-KFR	RECLOSING RELAYS								
								-KFS	SYNCHRONIZING DEVICES								
								-KFT	AUXILIARY TIME RELAYS, DELAY ELEMENTS								
								-KFZ	CONTROLLERS								
								-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER								
								-MAD	MOTOR FOR ELECTRICAL OPERATION OF DISCONNECTOR -QBD								
								-MAD1	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD1								
								-MAD2	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD2								
								-MAD9	MOTOR FOR ELECTRICAL OPERATION OF SWITCH-DISCONNECTOR -QBS								
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A	-MAE	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE						-QBH	MANUAL CIRCUIT-BREAKERS					
	-MAE1	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2						-QBL	LINKS					
	-MAE8	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS						-QBM	MINIATURE SWITCH-DISCONNECTORS					
	-MAS	MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING						-QBS	SWITCH-DISCONNECTORS					
	-MAT	MOTOR FOR ELECTRICAL OPERATION OF TRUCK RACKING-IN/OUT						-QBT	TRUCK					
	-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER						-QCA	EARTHING SWITCH FOR ELIMINATION OF INTERNAL ARCING					
	-MBC4	CLOSING RELEASE OF SWITCH-DISCONNECTOR -QBS						-QCE	EARTHING SWITCH					
	-MBO1	FIRST OPENING RELEASE OF CIRCUIT-BREAKER						-QCE1	EARTHING SWITCH OF BUSBARS (SINGLE BUSBAR SYSTEM) OR OF FRONT BUSBARS (DOUBLE BUSBAR SYSTEM)					
	-MBO2	SECOND OPENING RELEASE OF CIRCUIT-BREAKER						-QCE2	EARTHING SWITCH OF REAR BUSBARS (DOUBLE BUSBAR SYSTEM)					
	-MBO3	OPENING SOLENOID FOR OVERCURRENT RELEASE OF CIRCUIT-BREAKER OR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER						-QQ	CLUTCHES					
B	-MBO4	OPENING RELEASE OF SWITCH-DISCONNECTOR -QBS						-RAA	ANTIFERRORESONANCE RESISTOR					
	-MBU	UNDERVOLTAGE RELEASE OF CIRCUIT-BREAKER						-RAD	DIODES					
	-PFB	BLUE SIGNAL LAMPS						-RAI	INDUCTORS					
	-PFF	FLAG RELAYS						-RAR	RESISTORS					
	-PFG	GREEN SIGNAL LAMPS						-RF	FILTERS					
C	-PFR	RED SIGNAL LAMPS						-RLE1	ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING					
	-PFS	SHORT CIRCUIT INDICATORS						-RLE2	ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT					
	-PFV	VOLTAGE INDICATORS						-RLE3	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE					
	-PFV1	VOLTAGE INDICATORS ON FEEDER SIDE						-RLE4	ELECTROMECHANICAL LOCK PREVENTING THE DOOR OPENING OPERATION					
	-PFV2	VOLTAGE INDICATORS ON BUSBAR SIDE						-RLE5	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF DISCONNECTOR -QBD					
	-PFW	WHITE SIGNAL LAMPS						-RLE6	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD1					
	-PFX	CROSS INDICATORS, ELECTROMECHANICAL INDICATORS						-RLE7	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD2					
	-PFY	YELLOW SIGNAL LAMPS						-RLE8	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2					
	-PGA	AMMETERS						-RLE9	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF SWITCH-DISCONNECTOR -QBS					
	-PGC	COUNTERS						-RLE11	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD4					
D	-PGF	FREQUENCYMETERS						-RLE12	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD5					
	-PGH	HOURMETERS						-SFA	AMMETRIC SWITCHES					
	-PGI	PROTECTION AND CONTROL UNIT: HUMAN MACHINE INTERFACE						-SFC	CONTROL SWITCHES, CLOSING PUSH-BUTTONS					
	-PGI1	HUMAN MACHINE INTERFACE OF ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON CIRCUIT-BREAKER)						-SFL	LOCKING CONTACTS					
	-PGI2	HUMAN MACHINE INTERFACE FOR ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON SWITCHGEAR)						-SFO	OPENING PUSH-BUTTONS					
	-PGJ	ACTIVE ENERGY METERS						-SFR	RESET PUSH-BUTTONS					
	-PGK	REACTIVE ENERGY METERS						-SFS	SELECTOR SWITCHES					
	-PGM	MULTIFUNCTION INDICATORS						-SFT	TEST PUSH-BUTTONS					
	-PGP	POWER-FACTOR METERS						-SFU	UNLOCKING PUSH-BUTTONS					
	-PGQ	VARMETERS						-SFV	VOLTMETRIC SWITCHES					
E	-PGS	SYNCHRONOSCOPES						-TA	POWER TRANSFORMERS					
	-PGV	VOLTMETERS						-TB	CONVERTER					
	-PGW	WATTMETERS						-TB1	RECTIFIER FOR FIRST OPENING RELEASE					
	-PJ	ACOUSTICAL SIGNAL DEVICES (BELLS, SIRENS)						-TB2	RECTIFIER FOR SECOND OPENING RELEASE					
	-QAB	CIRCUIT-BREAKERS						-TB3	RECTIFIER FOR CLOSING RELEASE					
	-QAC	CONTACTORS (FOR POWER)						-TB4	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING					
	-QBD	DISCONNECTORS						-TB5	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT					
	-QBD1	FRONT BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						-TB6	RECTIFIER FOR UNDERVOLTAGE RELEASE					
	-QBD2	REAR BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						-TB7	ERECTIFIER FOR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER					
	-QBD3	FEEDER DISCONNECTORS (DOUBLE BUSBAR SYSTEM)												
F	-QBD4	TWO-WAY DISCONNECTOR												
	-QBD5	TWO-WAY DISCONNECTOR ON BUSBAR RISER												

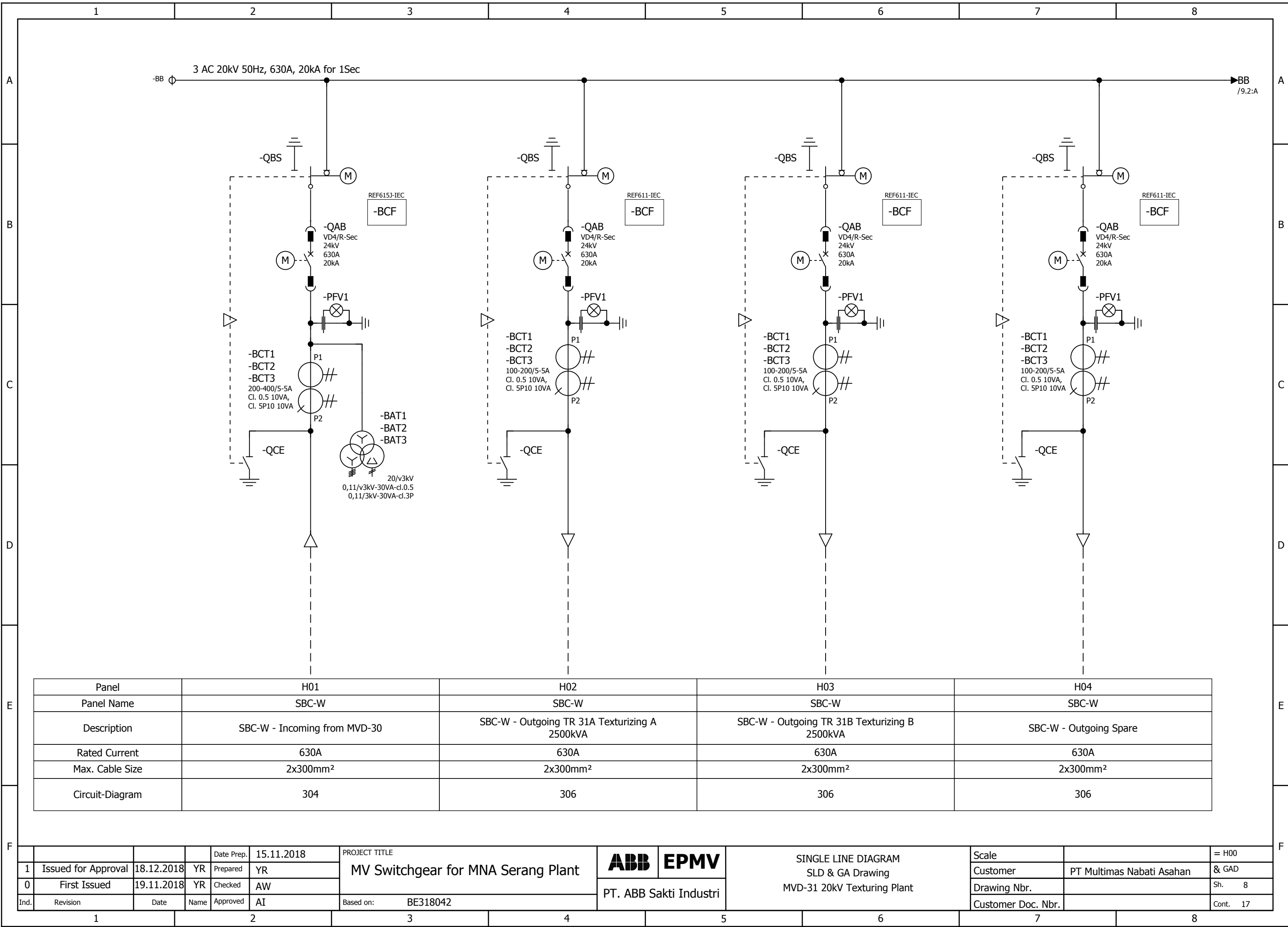
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1	Issued for Approval	18.12.2018	YR	Prepared	YR	MV Switchgear for MNA Serang Plant						Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW			PT. ABB Sakti Industri		Drawing Nbr.		Sh.	6	
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042				Customer Doc. Nbr.		Cont.	17	

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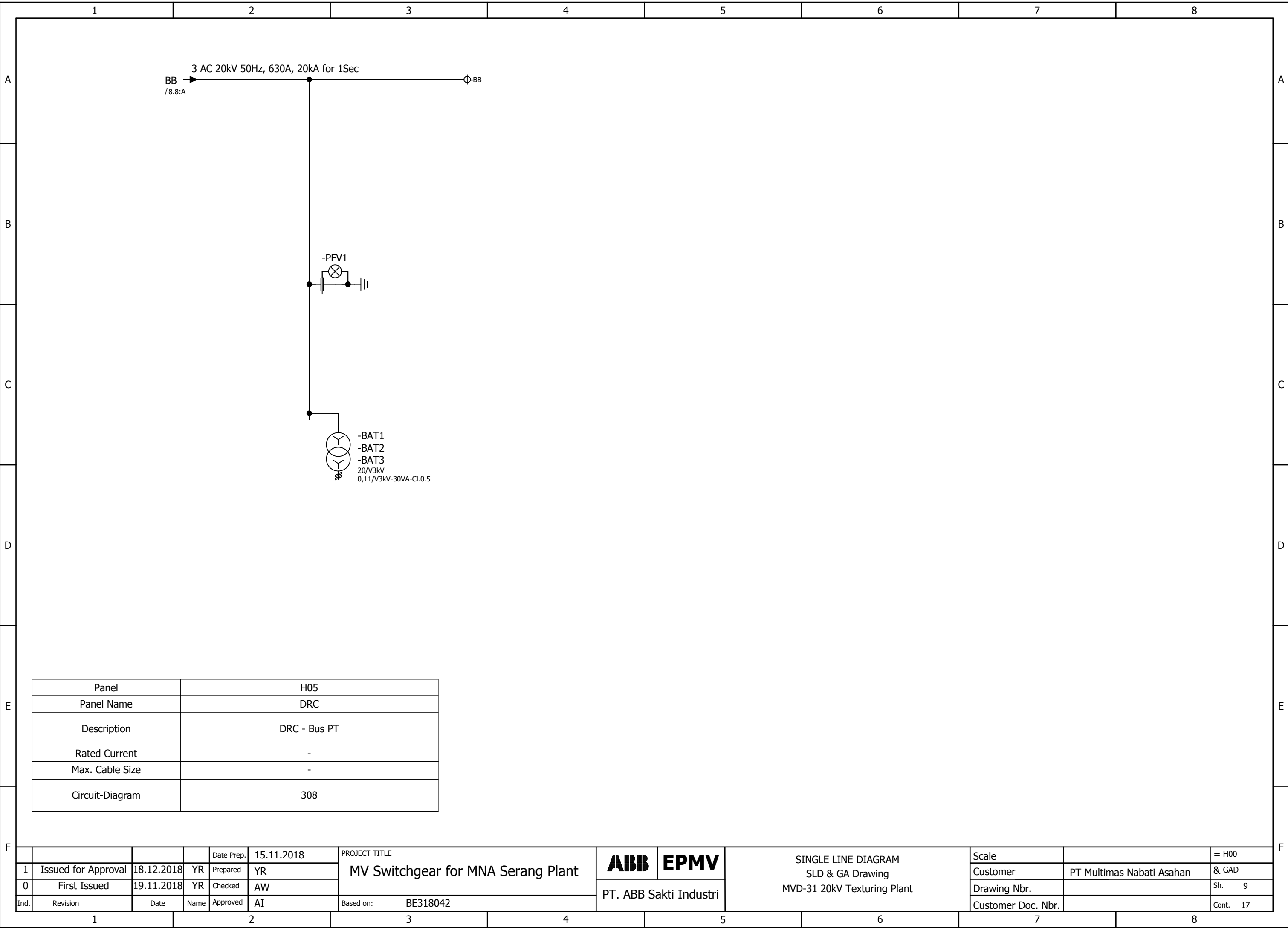
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1		2		3		4		5		6		7		8										
A	-TFA	ACTIVE POWER TRANSDUCERS						-XD11	TERMINAL BLOCK FOR INTERCONNECTIONS OF CONTROL CIRCUITS						A									
	-TFC	CURRENT TRANSDUCERS						-XD12	TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING															
	-TFF	FREQUENCY TRANSDUCERS						-XD13	TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR SWITCH-DISCONNECTOR OPERATION															
	-TFJ	ACTIVE ENERGY TRANSDUCERS						-XD14	TERMINAL BLOCK FOR INTERCONNECTIONS OF A.C. AUXILIARY CIRCUITS															
	-TFK	REACTIVE ENERGY TRANSDUCERS						-XD16	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS) OF VOLTAGE CIRCUITS															
	-TFM	MULTIFUNCTION TRANSDUCERS																						
B	-TFP	POWER-FACTOR TRANSDUCERS						-XD17	TERMINAL BLOCK FOR INTERCONNECTION OF MOD-BUS						B									
	-TFQ	REACTIVE POWER TRANSDUCERS						-XD18	TERMINAL BLOCK FOR INTERCONNECTIONS OF CURRENT CIRCUITS															
	-TFS	SIGNAL CONVERTERS						-XD19	TERMINAL BLOCK FOR SPECIAL USE															
	-TFV	VOLTAGE TRANSDUCERS						-XDJ	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF DISCONNECTORS -QBD															
	-WA	BUSBARS						-XDJ1	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR -QBS															
	-WBC	POWER CABLES						-XDM	SEALABLE TERMINAL BLOCK FOR MEASUREMENT															
C	-WE	EARTHING CABLES FOR L.V. INSTRUMENT COMPARTMENT.						-XDS	CURRENT SOCKETS						C									
	-WGC	CONTROL CABLES						-XDT	TERMINAL BLOCK FOR POSITION CONTACTS OF TRUCK AND OF CONNECTOR -XDB															
	-WGS	COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)						-XDU	CONNECTOR FOR ISOLATION OF CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK															
	-WH	OPTICAL FIBRES						-XDU1	TERMINAL BLOCK FOR CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK															
	-XDA	TERMINAL BLOCK FOR CIRCUITS OF CURRENT TRANSFORMERS						-XDV	TERMINAL BLOCK FOR CIRCUITS OF VOLTAGE TRANSFORMERS															
	-XDA1	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1						-XDV1	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L1															
D	-XDA2	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2						-XDV2	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L2						D									
	-XDA3	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3						-XDV3	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L3															
	-XDA5	CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER						-XDV4	CONNECTOR FOR CIRCUITS OF ANTIFERRORESONANCE RESISTOR															
	-XDB	CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS						-XDX	SUPPORT TERMINAL BLOCK															
	-XDB1	CONNECTOR FOR AUXILIARY CIRCUITS OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK)						-XE1	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, LEFT SIDE															
	-XDB2	CONNECTOR FOR ADDITIONAL AUXILIARY CIRCUITS OF CIRCUIT-BREAKER						-XE2	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, RIGHT SIDE															
E	-XDB10....-XDB89	CONNECTOR FOR CIRCUIT-BREAKER INTERNAL CIRCUITS						-XE5	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, VOLTAGE INDICATOR SIDE						E									
	-XDB7	CONNECTOR FOR TRUCK POSITION CONTACTS						-XE6	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, REAR BOTTOM SIDE															
	-XDB91	CONNECTOR FOR OPENING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR						-XE7	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, LEFT SIDE															
	-XDB92	CONNECTOR FOR CLOSING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR						-XE8	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, FRONT BOTTOM SIDE															
	-XDB93	CONNECTOR FOR POSITION SWITCHES OF SWITCH-DISCONNECTOR						-XE9	EARTHING TERMINAL BLOCK FOR TRANSFORMERS IN BUSBAR COMPARTMENT															
	-XDB95	CONNECTOR FOR CIRCUITS OF EARTHING SWITCH LOCKING MAGNET																						
F	-XDC	CUSTOMER TERMINAL BLOCK													F									
	-XDC1	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF CIRCUIT BREAKER																						
	-XDC2	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR																						
	-XDC3	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF PROTECTION RELAY AND MULTIFUNCTION UNIT																						
	-XDC4	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF VOLTAGE TRANSFORMERS																						
	-XDC5	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF MINIATURE CIRCUIT BREAKER																						
F	-XDE	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE													F									
	-XDE1	TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE																						
	-XDE2	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2																						
	-XDE3	TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2																						
	-XDF	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS																						
	-XDF1	CONNECTOR FOR ISOLATION OF CIRCUITS OF FRONT VENTILATOR																						
F	-XDF2	CONNECTOR FOR REAR VENTILATOR CIRCUITS													F									
	-XDF3	CONNECTOR FOR FRONT VENTILATOR CIRCUITS																						
	-XDH	TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)																						
	-XDI	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)																						
1		2		3		4		5		6		7		8										
					Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant		ABB	EPMV	REFERENCE DESIGNATIONS SLD & GA Drawing MVD-31 20kV Texturing Plant		Scale		= H00									
1	Issued for Approval	18.12.2018	YR	Prepared	YR								Customer	PT Multimas Nabati Asahan	& GAD									
0	First Issued	19.11.2018	YR	Checked	AW								Drawing Nbr.		Sh. 7									
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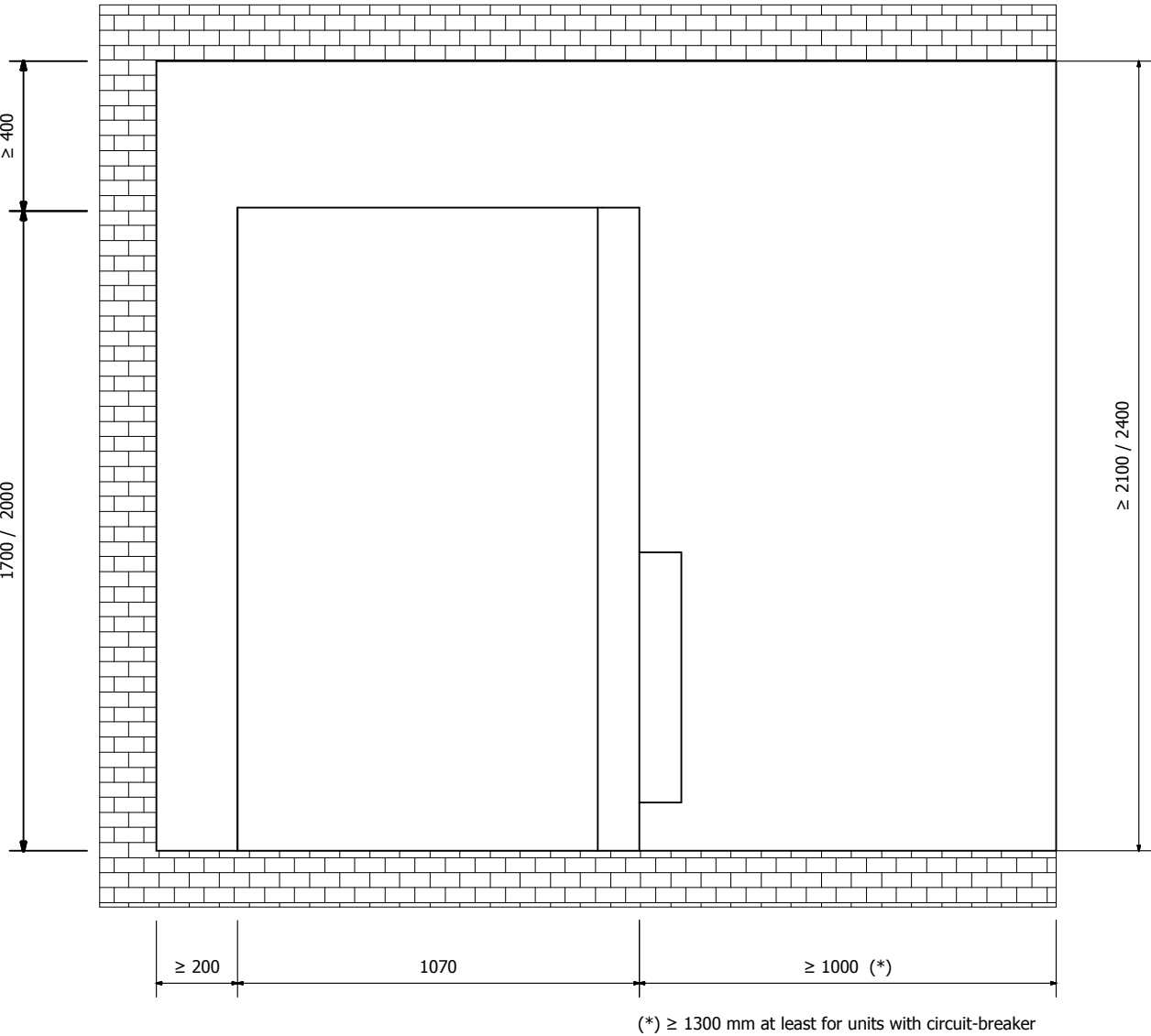
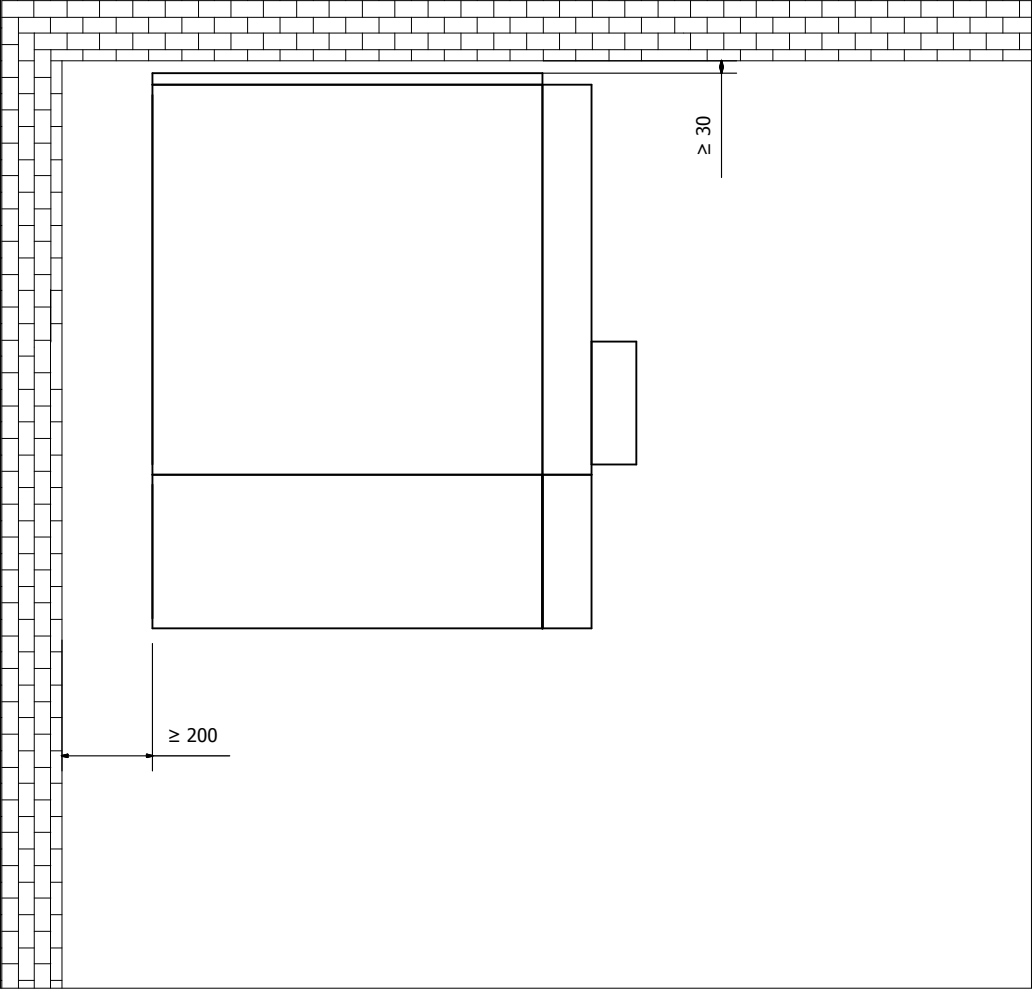
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RULES FOR INSTALLATION IN THE ROOM

MINIMUM DISTANCES TO BE RESPECTED DURING INSTALLATION



- IAC A-F SOLUTION UP TO 16kA 1sec
- ATTENTION : NO ACCESS TO REAR AND LATERAL OF THE SWG ROOM WHILE SWG IS IN SERVICE

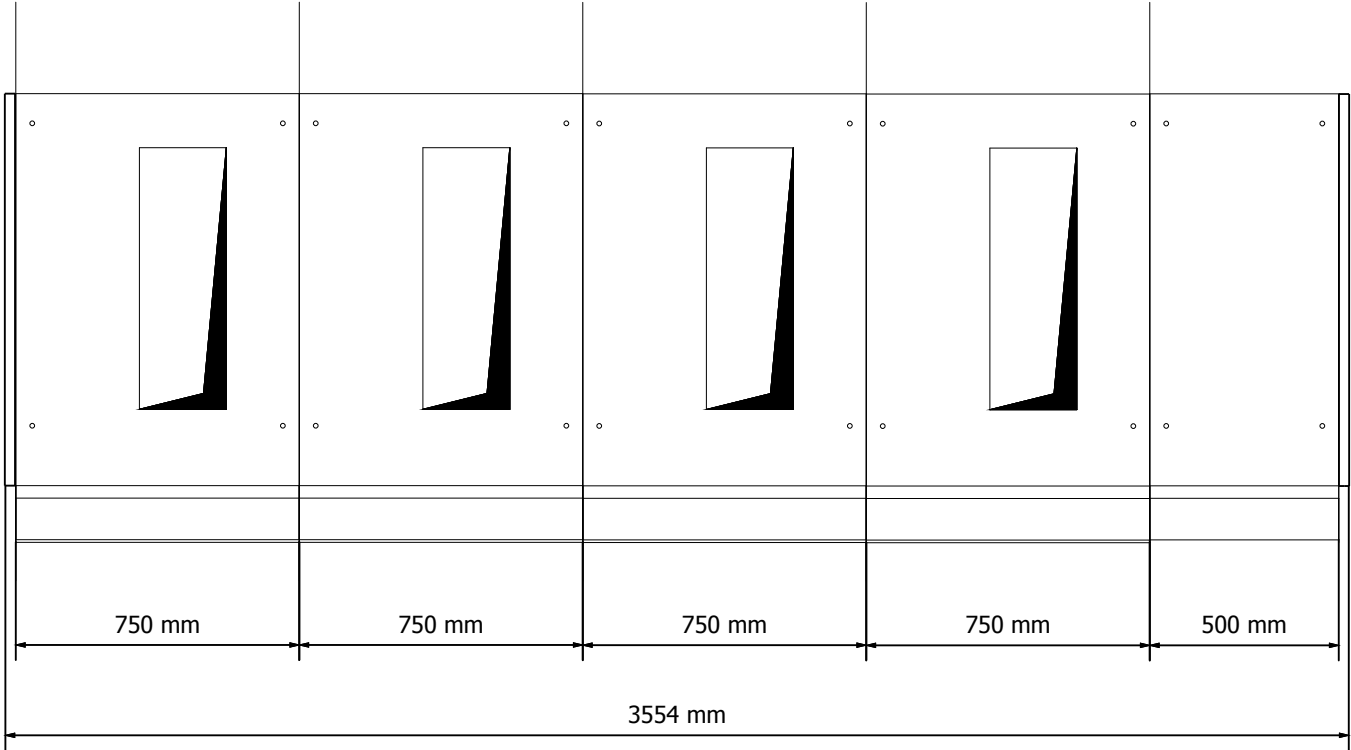
				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	INSTALLATION RULES SLD & GA Drawing MVD-31 20kV Texturing Plant	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					PT. ABB Sakti Industri	Customer	PT Multimas Nabati Asahan
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.				Sh. 10	
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	Customer Doc. Nbr.				Cont. 17	

Panel	H01	H02	H03	H04	H05
Panel Type	SBC-W	SBC-W	SBC-W	SBC-W	DRC
Description	SBC-W - Incoming	SBC-W - Outgoing TR 2500kVA	SBC-W - Outgoing TR 2500kVA	SBC-W - Outgoing	DRC - Bus PT

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	FRONT VIEW & GA SLD & GA Drawing MVD-31 20kV Texturing Plant	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					PT. ABB Sakti Industri	Customer	PT Multimas Nabati Asahan
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.				Sh. 11	
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	Customer Doc. Nbr.				Cont. 17	

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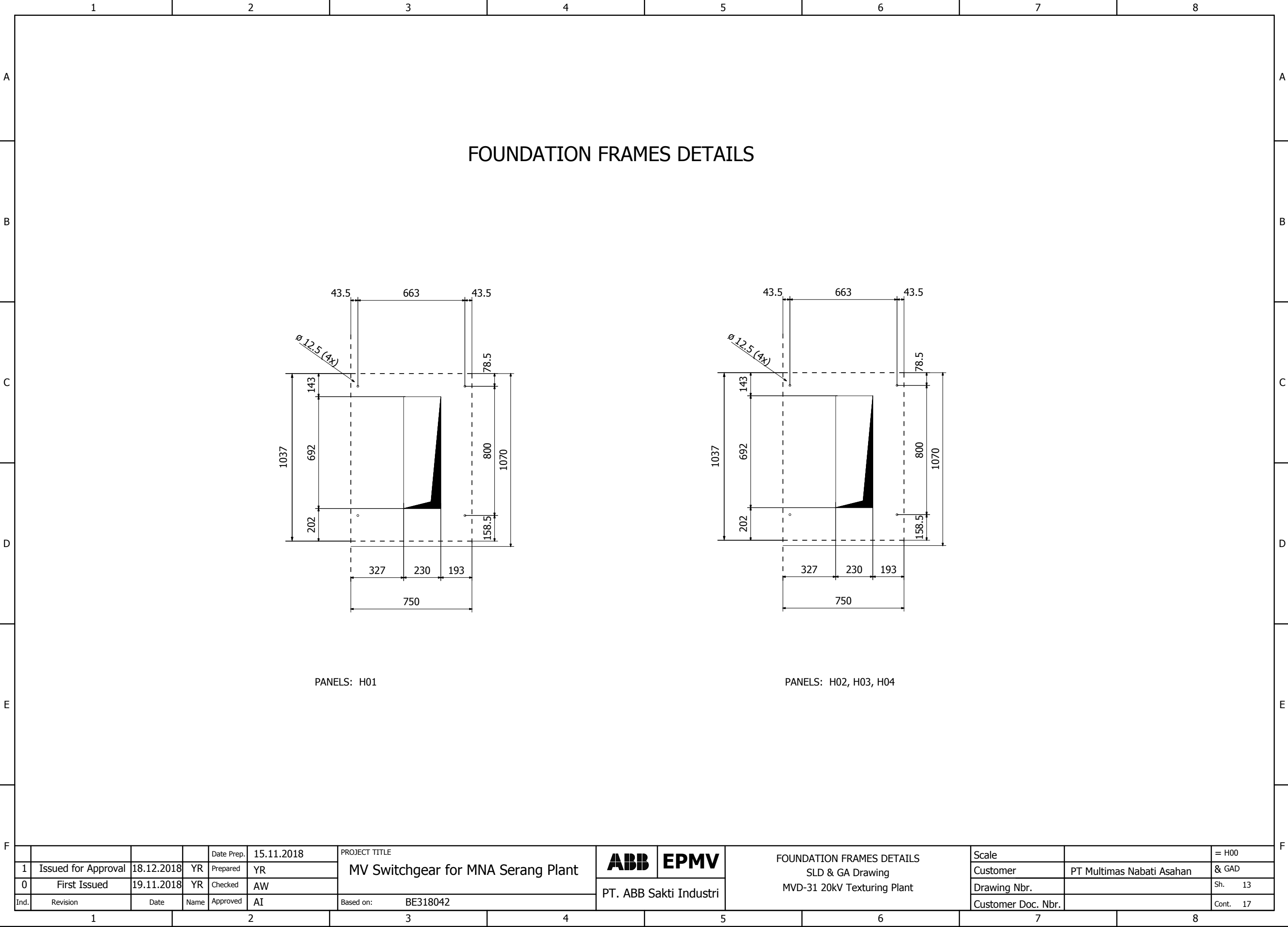
FOUNDATION FRAMES



Panel	H01	H02	H03	H04	H05
Panel Type	SBC-W	SBC-W	SBC-W	SBC-W	DRC
Description	SBC-W - Incoming	SBC-W - Outgoing TR 2500kVA	SBC-W - Outgoing TR 2500kVA	SBC-W - Outgoing	DRC - Bus PT

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	FOUNDATION FRAMES SLD & GA Drawing MVD-31 20kV Texturing Plant	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					Customer	PT Multimas Nabati Asahan	& GAD
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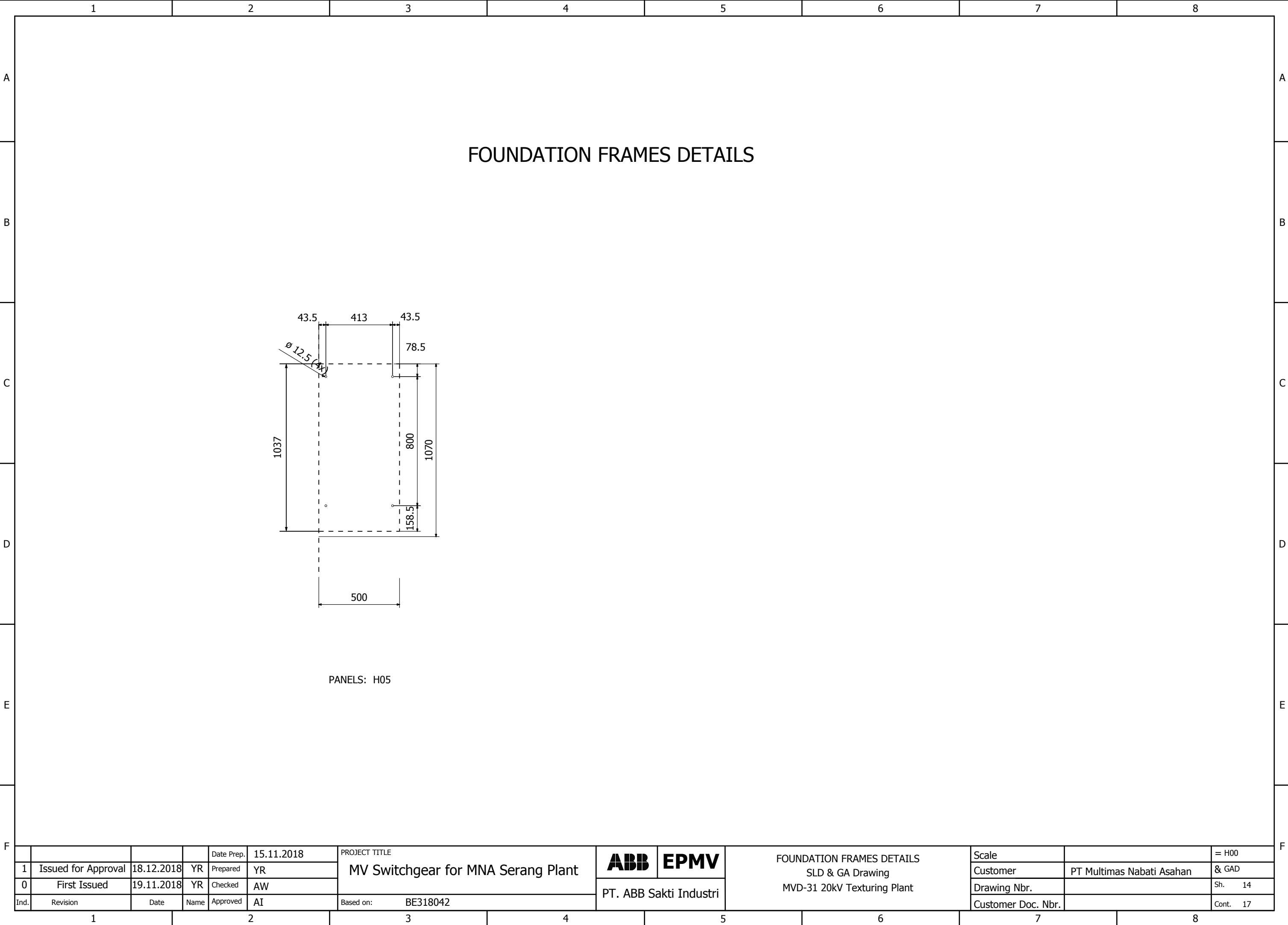
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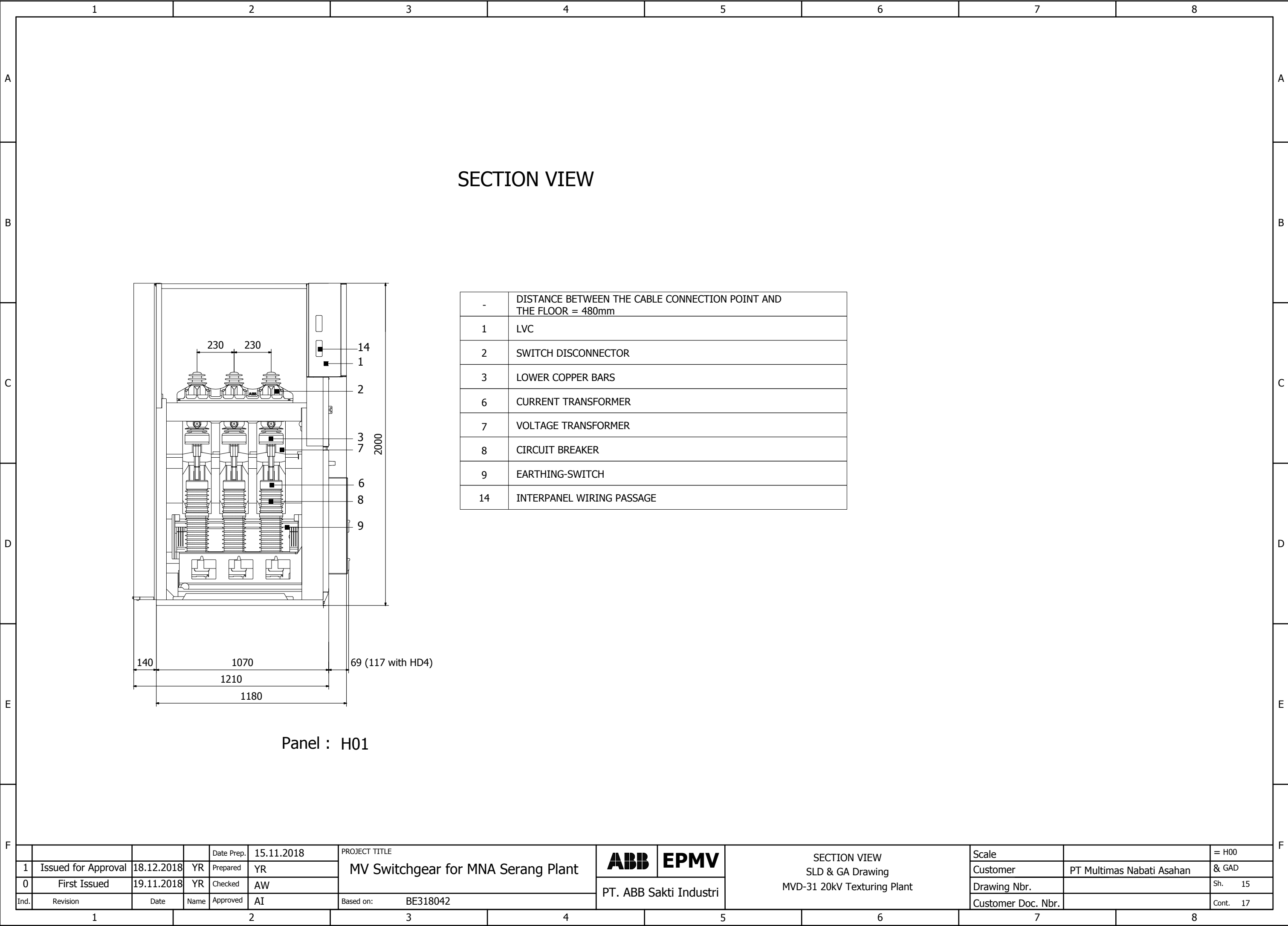
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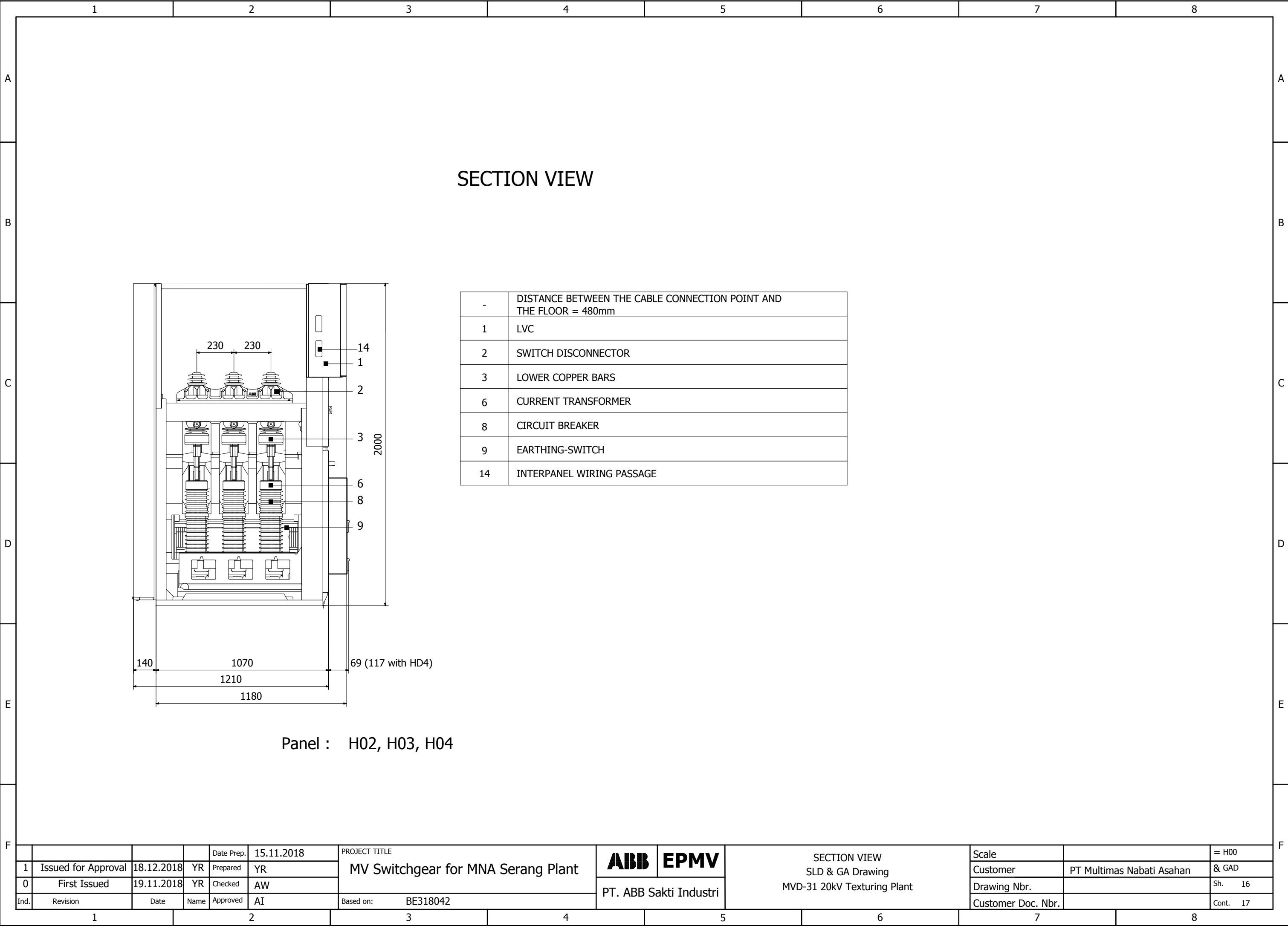
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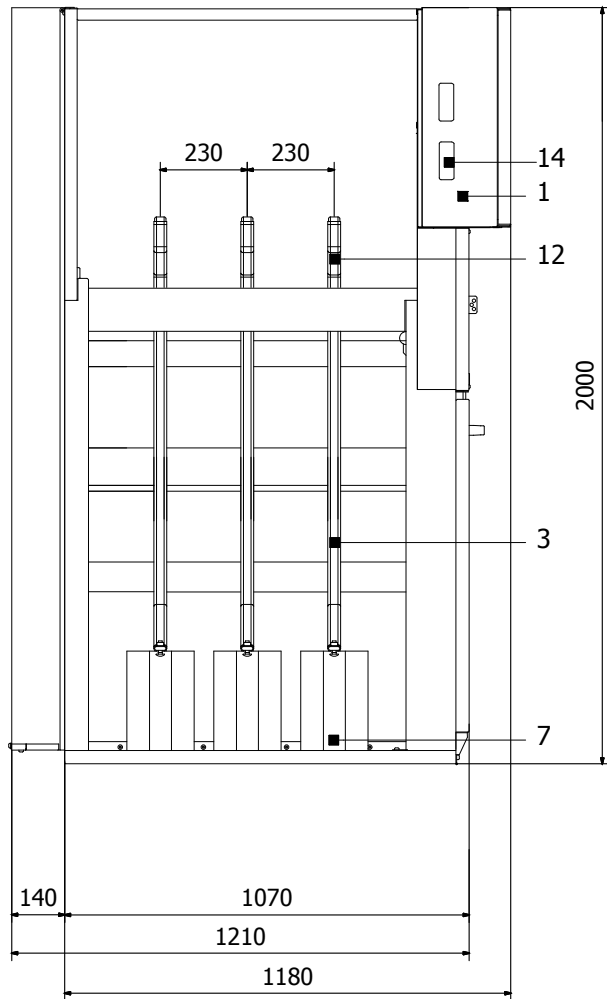


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SECTION VIEW



-	DISTANCE BETWEEN THE CABLE CONNECTION POINT AND THE FLOOR = 670mm
1	LVC
3	LOWER COPPER BARS
6	VOLTAGE TRANSFORMER
12	TOP COPPER BARS
14	INTERPANEL WIRING PASSAGE

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ABB EPMV